Toward an Arabic Punctuated Corpus: Annotation Guidelines and Evaluation

Wajdi Zaghouani[1] **and Dana Awad**[2]

[1]**Carnegie Mellon University in Qatar**
wajdiz@qatar.cmu.edu
[2]**The Lebanese University**
danaawad@hotmail.fr

Abstract

We present our effort to build a large scale punctuated corpus for Arabic. We illustrate in details our punctuation annotation guidelines designed to improve the annotation work flow and the inter-annotator agreement. We summarize the guidelines created, discuss the annotation framework and show the Arabic punctuation peculiarities. Our guidelines were used by trained annotators and regular inter-annotator agreement measures were performed to ensure the annotation quality. We highlight the main difficulties related to the Arabic punctuation annotation that arose during this project.

Keywords: Punctuation Annotation, Arabic Language, Guidelines, Evaluation

1. Introduction

Punctuation can be defined as the use of spacing and conventional signs to help the understanding of handwritten and printed texts. Punctuation marks are used to create sense, clarity and stress in sentences and also to structure and organize text.

Punctuation rules vary with language and register. Some punctuation aspects are of stylistic choices. For a language such as Arabic, punctuation marks are relatively a modern innovation since Arabic did not use punctuation and it was mainly introduced via translation and through borrowing from European languages as stated in (AlQinai, 2015) and therefore punctuation rules in Arabic are not consistently used although its becoming more popular in recent years. Asfour (2009) discussed this issue in his book on Arabic punctuation from which we cite the following quote:

> *"There is no doubt that Arabic has borrowed certain writing styles from foreign languages through translation, but the establishment of rules for Arabic punctuation does not depend on the various translations done by such and such translators, but mainly on what is commonly accepted by the Arabic writers."* Asfour (2009)

Attia et al. (2014) investigated the punctuation use in the parallel English/Arabic Gigaword corpus and found that around 10% of tokens in the English Gigaword corresponded to punctuation marks, while it is only 3% tokens in the in the Arabic counterpart.

From a Natural Language Processing (NLP) perspective, punctuation marks can be useful in the automatic sentence segmentation tasks, since sentence boundaries and phrase boundaries can be estimated based on punctuation marks. Moreover, as shown by a number of studies e.g., (Furui et al., 2004; Jones et al., 2003; Matusov et al., 2007), the absence of punctuation can be confusing both for humans and computers. Jones et al. (2003), for example, showed that sentence breaks are critical for text legibility. Furthermore, many NLP systems trained on well-formatted text often have problems when dealing with unstructured texts. Furui et al. (2004) showed that speech summarization improves when the sentence boundaries were available. Also, Matusov et al. (2007) showed that use of punctuation is beneficial for machine translation.

In order to build robust automatic punctuation systems, large scale manually punctuated corpora are usually needed. In this paper, we present our effort to build a large scale punctuated corpus for Arabic. We present our punctuation annotation guidelines designed to improve the inter-annotator agreement. Our guidelines were used by trained annotators and regular inter-annotator agreement measures were done to ensure the annotation quality during the lifespan of the project.

In the next sections, we briefly review the related work (Section 2), describe our corpus and the punctuation guidelines (Sections 3 and 4), and review our annotation procedure (Section 5), then we present the evaluation in Section 6.

2. Related Work

Large scale manually punctuated corpora are not yet widely available due to the high cost related to building such resources. We were able to locate some related works on punctuation annotation systems and resources such as the lightweight punctuation annotation system for the Arabic language built by Beeferman et al. (1998) to automatically insert intra-sentence punctuation into Arabic text. More recently, Attia et al. (2014) developed a punctuation errors correction tool based on a CRF (Conditional Random Fields) classifier, and Mubarak and Darwish (2014) used two approaches to restore punctuation marks in Arabic text, in the first approach they used a simple statistical word-based system, and in the second approach a conditional random fields (CRF) sequence labeler that tries to recover punctuation based on Part-of-Speech (POS) of the current word and also of the two previous and following words. They used also word sequences such as the current word, the previous or the next word in the sentence.

For English, Spitkovsky et al. (2011) showed how punctuation can be used to improve unsupervised dependency parsing. They presented a linguistic analysis to confirm the connection between English punctuation and phrase boundaries in the Penn Treebank. The Arabic punctuated corpus we present in this paper is a part of a large scale manually error annotated project described in the next section.

3. Corpus Description

In this work, we describe a 2 million words corpus developed for the Qatar Arabic Language Bank (QALB) project, a large-scale annotation effort created to deliver a manually corrected corpus of errors including punctuation errors for a variety of Arabic texts (Zaghouani et al., 2014b; Zaghouani et al., 2015; Zaghouani et al., 2016). The overreaching goal of the punctuation annotation sub-task in this project is twofold: correct the existing punctuation found in text, then add the missing necessary punctuation when needed. The corpus presented contains a variety of text genres and sources: (a) user comments on news websites (L1 corpus), (b) native speaker essays (L1 corpus) (c) non-native speaker essays (L2 corpus), (d) machine translation output (MT corpus) as shown in Table 1 . The *user comments* portion is 1.5 million words corpus collected from AlJazeera.Net news.[1] The comments are selected from the related news stories. The *native student essays corpus* includes 151,000 words extracted from the Arabic Learners Corpus (ALC) Alfaifi and Atwell (2012) and 66,000 words from the University Students Essays Corpus (Alkanhal et al., 2012).

The *non-native student essays corpus* has 131,000 words selected from the Arabic Learners Corpus (ALC) by Alfaifi and Atwell (2012) and 51,000 words from the Arabic Learners Written Corpus (ALWC) compiled by Farwaneh and Tamimi (2012) . The data is organized by the students level (beginner, intermediate, advanced), learner type (L2 vs. heritage),[2] and essay type (description, narration, instruction).

Finally, the *machine translation output corpus* was collected from English news articles covering 100,000 words extracted from the collaborative journalism Wikinews website[3]. The corpus includes 520 articles with an average of 192 words per article. The original English files were in HTML format and later on exported to a UTF-8 plain Text standard format so it can be uploaded in the annotation tool. Afterwards, the collected corpus was automatically translated from English to Arabic using the Google Translate API paid service [4].

4. Punctuation Annotation Guidelines

Punctuation errors are estimated to constitute 40% of the errors in the QALB corpus, that is 10 times higher than the 4% of punctuation errors found in the English data used in CoNLL 2013 Shared Task on English Grammatical Error Correction (Ng et al., 2013). As mentioned earlier in this article, punctuation use in English or French language is guided by a series of grammar-related rules, while in other languages such as Arabic, punctuation is a recent innovation as premodern Arabic did not use punctuation (Zaki, 1995). According to Awad (2013), there is an inconsistency in the punctuation rules and use in Arabic, and omitting the punctuation marks is very frequent. To create our guidelines, we use the Arabic standard punctuation rules commonly used today and described in Awad (2013).

The annotation guidelines typically document the core of the annotation policy. Our punctuation guidelines focus on the types of punctuation errors that are targeted and they describe how the correction should be done and also when new punctuation marks should be added. Many annotated examples are provided in the

[1] We thank AlJazeera.Net for granting the rights to use the user comments from their website in our corpus.

[2] Heritage learner is generally used to describe a person learning a language and who has some proficiency in or he is culturally connected to that language through country of origin or family.

[3] https://en.wikinews.org

[4] https://cloud.google.com/translate

Corpus Type	Source	Size
News Comments (L1)	Aljazeera.Net	1500
Native Students Essays (L1)	Arabic Learners Corpus (Alfaifi and Atwell, 2012)	151
Native Students Essays (L1)	University Students Essays (Alkanhal et al., 2012)	66
Non-Native Students Essays (L2)	Arabic Learners Corpus (Alfaifi and Atwell, 2012)	131
Non-Native Students Essays (L2)	Arabic Learners Written Corpus (Farwaneh and Tamimi, 2012)	51
Machine Translation Output	Wikinews (English-Arabic MT)	100

Table 1: Corpus types and sources used in the QALB project. The size is displayed in K-words.

guidelines to illustrate the various annotation rules and exceptions.

Since the Arabic punctuation rules are not always clearly defined, we adopted an iterative approach for developing the guidelines, which includes multiple revisions and updates needed for the different rounds of of annotation in order to reach a consistent set of instructions. In order to help the annotators deal with some complex punctuation rules, we wrote a summary of the most common punctuation marks rules in Arabic as an appendix to the guidelines.

Furthermore, we instructed the annotators to convert any Latin punctuation sign to the equivalent Arabic sign when applicable. In particular, we instructed them to use always the Arabic comma ، the Arabic Semicolon ؛ and the Arabic question marks ؟ instead of the English equivalent punctuation marks. The punctuation annotation guidelines are published as a technical report (Zaghouani et al., 2014a) and they are currently available for download from the Qatar Arabic Language Bank project web page.[5]

The rules of punctuation vary with the language and the register. Moreover, some aspects of punctuation use vary from author to author, and can be considered a stylistic choice. Punctuation errors are especially present in student essays and online news comments. This is mainly due to the fact that some punctuation mark rules are still not clearly defined in Arabic writing references as explained previously. Table 2 shows an example of two punctuation errors and their corrections.

We created a set of simple rules for correcting punctuation and adding the missing ones. Below, we list a portion of the punctuation guidelines to illustrate some of the most important punctuation marks used in Arabic.

The Comma: The Arabic comma correction rules state the following four uses as valid: (1) to separate coordinated and main-clause sentences, usually between short sentences, in order to specify that there is a continuation in the topic; (2) during enumeration to avoid repetition; (3) to provide an explanation or a definition of the previous word; and (4) to separate between parts of the conditional sentences. Furthermore, while English places commas between adjectives of equal weight, Arabic uses zero punctuation or the optional coordination conjunction wa و 'and'. In other cases, a lexical substitute is inserted instead of the comma. For example the Arabic expression: إلا أن 'except that' can be used to replace the comma.

Error	أحب السفر كل، صيف ولكن لن أسافر هذا العام
Edit	أحب السفر كل صيف ولكن لن أسافر هذا العام.
English	I like to travel every summer but I won't this year.

Table 2: Example of two punctuation errors. A comma is used in the wrong place and the sentence does not end with a period.

The Period: The Arabic period or full stop is used at the end of declarative sentences and it is similar to the common use in other languages such in English.

Semi-Colon: In Arabic the semicolon is called Fasila Manquta, literally meaning a dotted comma and is written inverted. In general, Arabic semi-colons are used to indicate that what follows the semicolon is explaining, elaborating, or justifying what precedes it. It can also be used in the following two cases: It can be used between two phrases, in which the first phrase causes the second. And it can be used in two phrases, where the second is a reason for the first. In our guidelines and according to Newmark (1984), Arabic semicolon is also used between two parallel sentences bearing similarity or contrast. In a similar way to the Arabic comma, the Arabic semicolon can be substituted to a coordination conjunction wa و 'and'.

The Colon: The most common use of the colon is to inform the reader that what follows the colon proves, explains, or lists elements of what preceded it. Be-

[5]http://nlp.qatar.cmu.edu/qalb/

low is a restricted list of when the colon may be used during the annotation.

1. Following a dialogue or a conversation as in سألته: من أين لك هذا ؟ 'I asked him: from where you got this?'.

2. To enumerate the classes or the types of related objects as in the following example أيام الدهر ثلاثة: يوم مضى.... 'the days of life are of three types : a day that passed...'.

3. Following some specific Arabic expressions such as التالية 'the next' and ما يلي 'what comes next'.

4. In the case of citations introduced in the text.

The Question mark: This punctuation mark replaces the period at the end of an interrogative sentence. Arabic question mark may be used when there is an Arabic interrogative particle in the sentence such as من 'who', متى 'when', كيف 'how' and أين 'where' as in the following examples: من قال كذا؟ 'Who said such and such?' متى تصل؟ 'When will you arrive?' كيف تعمل؟ 'How it works?' أين أخوك؟ 'Where is your brother?'.

The Exclamation Mark: The exclamation mark is usually used after an interjection or exclamation to indicate strong feelings or high volume (shouting), and often marks the end of a sentence. In Arabic, the exclamation mark may be used in a similar way to the English as in the following example: كم هذا رائع! 'How wonderful!'.

5. Annotation Procedure

The lead annotator acts as the annotation work-flow manager in this project. He frequently evaluates the quality of the annotation, monitor and reports on the annotation progress. A clearly defined protocol is set, including a routine for the annotation job assignment. The lead annotator is also responsible for the corpus selection and normalization process, besides the annotation of the gold standard files to be used to compute the Inter-Annotator Agreement (IAA) portion of the corpus needed to control the quality of the annotation. The annotators in this project are five Arabic native speakers university graduates with good Arabic language background. To ensure the annotation consistency, an extensive training phase for each annotator was conducted. During the training phase, various meetings were organized to have the annotators read the guidelines in groups and practice various annotation tasks before discussing issues raised during the training.

Afterwards, the annotator's performance is closely monitored during the initial period, before allowing the annotator to join the official annotation production phase. Furthermore, a dedicated on-line discussion group is frequently used by the annotation team to keep track of the punctuation questions and issues raised during the annotation process, this mechanism, proved to help the annotators and the lead annotator to have a better communication.

The annotation itself is done using a web annotation framework built originally for the manual correction annotation of errors in the QALB project (Obeid et al., 2013). This project framework includes two major components:

1. The annotation management interface used to assist the lead annotator in the general work-flow process. The interface allows the user to upload, assign, monitor, evaluate and export annotation tasks.

2. The annotation interface is the actual annotation tool (Figure 1), used by the annotators to do the manual correction of the Arabic text and to add the missing punctuation or correct the existing ones. As shown in Figure 1, edited words and punctuation marks are highlighted in blue.

All the annotation history is recorded in a database and can be exported to an XML export file to keep a trace of the entire correction actions for a given file as shown in Figure 2.

6. Evaluation

To evaluate the punctuation annotation quality, we measure the inter-annotator agreement (IAA) on randomly selected files to ensure that the annotators are consistently following the annotation guidelines. A high annotation agreement score is a good indicator of the data quality.

The IAA is measured by averaging WER over all pairs of annotations to compute the AWER (Average Word Error Rate). In this evaluation, the WER measures the punctuation errors against all existing punctuation marks in the text. The IAA results shown in Table 3 are computed over 10 files from each corpus (30 files and 4,116 words total) annotated by at least three different annotators. As seen in the table 3, we noticed an improvement over time of the agreement, since the project started with the L1 corpus, then the L2 corpus and finally the MT corpus. This can be explained in a way by the experience gained overtime by the annotators during the project and also by the frequent guidelines updates done to simplify the punctuation guide-

Figure 1: The annotation interface. Edited words are highlighted in blue.

```
<DOCUMENT>
  <ACTION_HISTORY>
    <ACTION actionType="edit" annotatorID="23" newText="," passNum="1" tokenID="43" />
    <ACTION actionType="edit" annotatorID="23" newText="." passNum="1" tokenID="85" />
    <ACTION actionType="edit" annotatorID="23" newText="!" passNum="1" tokenID="91" />
    <ACTION actionType="edit" annotatorID="23" newText=":" passNum="1" tokenID="93" />
    <ACTION actionType="edit" annotatorID="27" newText=";" passNum="2" tokenID="7" />
    <ACTION actionType="edit" annotatorID="27" newText="؟" passNum="2" tokenID="18" />
      </ACTION_HISTORY>
</DOCUMENT>
```

Figure 2: Extract of output file showing some punctuation correction action history

lines. Moreover, we noticed a much higher agreement with the MT corpus. A closer analysis revealed that this was caused by a much less occurrence of the comma in the MT corpus as compared to L1 and L2 corpus as seen in Table 4. Indeed the comma punctuation mark is not always clearly marked in many cases and it can be substituted by the Arabic conjunction wa و 'and'. Furthermore, the MT corpus has a much higher occurrence of the period mark which is considered easier to edit. Overall, the results obtained showed that the annotators are consistently following the punctuation guidelines provided to them.

Corpus	Average IAA	Average WER
L1 Corpus	89.84	10.16
L2 Corpus	91.9	8.1
MT Corpus	93.78	6.22

Table 3: Average percent Inter-Annotator Agreement (IAA) and the Average WER (AWER) obtained for the three corpora.

Punctuation Mark	L1	L2	MT
Comma	44.77%	45%	38%
Period	37%	40%	50%
Question Mark	8.79%	7.38%	4.50%
Exclamation Mark	4.89%	4%	3%
Colon	2.56%	2.12%	2.50%
Semi Colon	2%	1.50%	2%
Total	100%	100%	100%

Table 4: Distribution of each punctuation mark in the three corpora.

7. Conclusions

We presented our method to create an Arabic manually punctuated corpus, including the writing of the guidelines as well as the annotation procedure and the quality control method used to verify the annotation quality. We showed that there is a high variety in the use of punctuation in Arabic texts and despite the existence of punctuation rules, the use of punctuation in Arabic is highly individual and it depends on the style of the author who may define his/her own use of punctuation. The specific punctuation annotation

guidelines presented in this paper helped to reduce the variations in the punctuation use during the annotation experiment.

The created resource has been already freely released for the research community during the related shared tasks we organized in recent years (Mohit et al., 2014; Rozovskaya et al., 2015). We hope to receive feedback from the community on the usefulness and potential related applications of the resource.

Acknowledgements

We would like to thank the anonymous reviewers for their valuable comments and suggestions. We also thank all our dedicated annotators: Noor Alzeer, Hoda Fathy, Hoda Ibrahim, Anissa Jrad and Jihene Wafi. We thank Nizar Habash, Behrang Mohit, Kemal Oflazer and Ossama Obeid for their valuable contribution in this project. This publication was made possible by grants NPRP-4-1058-1-168 from the Qatar National Research Fund (a member of the Qatar Foundation). The statements made herein are solely the responsibility of the authors.

8. References

Alfaifi, A. and Atwell, E. (2012). Arabic Learner Corpora (ALC): A Taxonomy of Coding Errors. In *The 8th International Computing Conference in Arabic (ICCA 2012)*, Cairo, Egypt.

Alkanhal, M. I., Al-Badrashiny, M. A., Alghamdi, M. M., and Al-Qabbany, A. O. (2012). Automatic Stochastic Arabic Spelling Correction With Emphasis on Space Insertions and Deletions. *IEEE Transactions on Audio, Speech, and Language Processing*, 20(7):2111–2122.

AlQinai, J. (2015). Mediating punctuation in english arabic translation. *Journal of Applied Linguistics and Professional Practice*, 5(1).

Asfour, M. (2009). *at-Tarqim fi l-lugati al-arabiyya*. Dar al-Baraka, Amman.

Attia, M., Al-Badrashiny, M., and Diab, M. (2014). Gwu-hasp: Hybrid arabic spelling and punctuation corrector. In *Proceedings of the EMNLP 2014 Workshop on Arabic Natural Langauge Processing (ANLP)*, pages 148–154.

Awad, D. (2013). La ponctuation arabe : histoire et règles.

Beeferman, D., Berger, A., and Lafferty, J. (1998). Cyberpunc: a lightweight punctuation annotation system for speech. In *Acoustics, Speech and Signal Processing, 1998. Proceedings of the 1998 IEEE International Conference on*, volume 2, pages 689–692 vol.2, May.

Farwaneh, S. and Tamimi, M. (2012). Arabic Learners Written Corpus: A Resource for Research and Learning. *The Center for Educational Resources in Culture, Language and Literacy.*

Furui, S., Kikuchi, T., Shinnaka, Y., and Hori, C. (2004). Speech-to-text and speech-to-speech summarization of spontaneous speech. *IEEE Transactions on Speech and Audio Processing*, 12(4):401–408.

Jones, D. A., Wolf, F., Gibson, E., Williams, E., Fedorenko, E., Reynolds, D. A., and Zissman, M. A. (2003). Measuring the readability of automatic speech-to-text transcripts. In *INTERSPEECH*. ISCA.

Matusov, E., Hillard, D., Magimai-doss, M., Hakkani-tur, D., Ostendorf, M., and Ney, H. (2007). Improving speech translation with automatic boundary prediction. In *Proceedings of Interspeech*, pages 2449–2452.

Mohit, B., Rozovskaya, A., Habash, N., Zaghouani, W., and Obeid, O. (2014). The first qalb shared task on automatic text correction for arabic. In *Proceedings of the EMNLP Workshop on Arabic Natural Language Processing*, page 39.

Mubarak, H. and Darwish, K. (2014). Automatic correction of arabic text: a cascaded approach. In *Proceedings of the EMNLP 2014 Workshop on Arabic Natural Langauge Processing (ANLP)*, pages 148–154.

Newmark, P. (1984). *Approaches to Translation*. Oxford.

Ng, H. T., Wu, S. M., Wu, Y., Hadiwinoto, C., and Tetreault, J. (2013). The conll-2013 shared task on grammatical error correction. In *Proceedings of the Seventeenth Conference on Computational Natural Language Learning: Shared Task*, pages 1–12, Sofia, Bulgaria, August. Association for Computational Linguistics.

Obeid, O., Zaghouani, W., Mohit, B., Habash, N., Oflazer, K., and Tomeh, N. (2013). A Web-based Annotation Framework For Large-Scale Text Correction. In *The Companion Volume of the Proceedings of IJCNLP 2013: System Demonstrations*, Nagoya, Japan, October.

Rozovskaya, A., Bouamor, H., Habash, N., Zaghouani, W., Obeid, O., and Mohit, B. (2015). The second qalb shared task on automatic text correction for arabic. In *Proceedings of the ACL-IJCNLP Workshop on Arabic Natural Language Processing*, page 26.

Spitkovsky, V. I., Alshawi, H., and Jurafsky, D. (2011). Punctuation: Making a point in unsu-

pervised dependency parsing. In *Proceedings of the Fifteenth Conference on Computational Natural Language Learning*, CoNLL '11, pages 19–28, Stroudsburg, PA, USA. Association for Computational Linguistics.

Zaghouani, W., Habash, N., and Mohit, B. (2014a). The qatar arabic language bank guidelines. Technical Report CMU-CS-QTR-124, School of Computer Science, Carnegie Mellon University, Pittsburgh, PA, September.

Zaghouani, W., Mohit, B., Habash, N., Obeid, O., Tomeh, N., Rozovskaya, A., Farra, N., Alkuhlani, S., and Oflazer, K. (2014b). Large scale arabic error annotation: Guidelines and framework. In *International Conference on Language Resources and Evaluation (LREC 2014)*.

Zaghouani, W., Habash, N., Bouamor, H., Rozovskaya, A., Mohit, B., Heider, A., and Oflazer, K. (2015). Correction annotation for non-native arabic texts: Guidelines and corpus. In *Proceedings of the Association for Computational Linguistics Fourth Linguistic Annotation Workshop*, pages 129–139.

Zaghouani, W., Habash, N., Obeid, O., Mohit, B., Bouamor, H., and Oflazer, K. (2016). Building an arabic machine translation post-edited corpus: Guidelines and annotation. In *International Conference on Language Resources and Evaluation (LREC 2016)*.

Zaki, A. (1995). at-tarqim wa alamatu-hu fi al-lugati al-arabiyya, maktabat al-matbuat al islamiyya bi halab.